Chapter 6 • Trigonometry

Solved Exercise 6.6

1. A flagpost is observed at 60° from 40 m away. Find its height h.

$\tan 60° = \frac{h}{40}$

$\Rightarrow h = 40\sqrt{3}$.

69.28 m approximately.

2. An isosceles triangle has vertical angle 120° and base 10 cm. Find its altitude.

The altitude bisects the vertical angle and base. $\tan 60° = \frac{5}{h}$

$h = \frac{5}{\sqrt{3}} = 2.89$ cm approximately.

3. A 72 m tree is observed from 100 m away. Find the angle of elevation.

$\theta = \tan^{-1}(\frac{72}{100})$

$= 35.753...°$.

35.7° approximately.

4. A ladder at 60° reaches 10 m up a wall. Find its length L.

$\sin 60° = \frac{10}{L}$

$\Rightarrow L = \frac{10}{\frac{\sqrt{3}}{2}} = \frac{20}{\sqrt{3}}$

11.55 m approximately.

5. From a 150 m lighthouse, the angle of depression to a ship is 60°. Find horizontal distance d.

The elevation angle at the ship is also 60°. $\tan 60° = \frac{150}{d}$

$d = \frac{150}{\sqrt{3}} = 86.60$ m approximately.

6. A pole's elevation changes from 15° to 30° after walking 100 m toward it. Find its height h.

Initial distance$= h \cot 15°$

final distance$= h \cot 30°$.

$100 = h(\cot 15° - \cot 30°)$. Since $\cot 15° = 2 + \sqrt{3}$ and $\cot 30° = \sqrt{3}$, the difference is 2.

h=50 m exactly (about 49.98 m with rounded tables).

7. A 300 m tower casts a 450 m shadow. Find the Sun's elevation.

$\theta = \tan^{-1}(\frac{300}{450})$

$= \tan^{-1}(\frac{2}{3})$.

33.69° approximately.

8. A cliff's elevation changes from 25° to 45° after walking 100 m toward it. Find height h.

At 45°, near distance = h. Initial distance

$= h + 100$.

$\tan 25° = \frac{h}{h+100}$

$\Rightarrow h(1 - \tan 25°) = 100 \tan 25°$.

$h = \frac{100 \tan 25°}{1 - \tan 25°} = 87.37$ m≈ 87.4 m.

9. From a 300 m hill, depression angles to the nearer and farther riverbanks are 70° and 50°.

Near distance $= 300 \cot 70°$

$= 109.19$ m

far distance $= 300 \cot 50°$

$= 251.73$ m.

River width=far distance−near distance.

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Width≈142.54 m (142.5 m); nearer bank is 109.19 m from the hill’s foot.

10. A kite has 120 m of stretched string at elevation 50°. Find its height above the hand.

h=120sin50°.

91.92 m approximately.

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